Printed Pages: 02 **Sub Code: KME-303**

Paper Id: 233653 **Roll No.** | | | | | | | | | | |

B Tech.
(SEM III) THEORY EXAMINATION 2022-23
MATERIALS ENGINEERING

Time: 3 Hours ***Total Marks: 100***

Note: Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt *all* questions in brief. **2 x 10 = 20**

(a) Define Unit Cell.

(b) Define Malleability.

(c) What do you mean by Fatigue? Explain.

(d) Explain Non-Destructive Testing.

(e) Differentiate between alloy and ore.

(f) Explain Degree of Freedom used in Phase Diagram.

(g) Describe the Heat Treatment.

(h) Explain the need or purpose of Heat Treatment.

(i) Mention the percent of carbon (range) used in Tool Steel.

(j) Differentiate between Martensite and Cementite forms.

SECTION B

2. Attempt any *three* of the following: **10x3=30**

(a) Explain the Crystal Structure (System) in detail.

(b) Elaborate the ductile and brittle failure mechanism.

(c) Discuss the Iron-Carbon Phase diagram in detail with neat sketch.

(d) Describe the process of Annealing and Normalizing and its effect on different properties.

(e) Mention the properties of Stainless Steel and Tool Steel. Also discuss their utilization for engineering materials.

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QP23DP2_311 | 01-04-2023 13:28:51 | 115.243.172.58

SECTION C

3. Attempt any *one* part of the following: 10x1=10

(a) Discuss the Line Imperfections with the help of neat sketches.
(b) Explain the Load-Deformation diagram for a brittle material.

4. Attempt any *one* part of the following: 10x1=10

(a) Name the various types of Static Failure Theories. Explain any one in detail.
(b) Discuss the SN Curve in detail. Explain typical S-N curve for ferrous and non-ferrous alloys.

5. Attempt any *one* part of the following: 10x1=10

(a) What do you mean by Solid Solution? Explain the various types of Solid Solutions.
(b) Describe various types of micro-constituents of Eutectic Steel (at STP).

6. Attempt any *one* part of the following: 10x1=10

(a) What is Tempering? Explain its need and types also.
(b) Explain Case Hardening? Mention its types.

7. Attempt any *one* part of the following: 10x1=10

(a) Discuss the constituents and properties of Nickel based Super alloys.
(b) Discuss the constituents and properties of Titanium Alloys.

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